

Systems Design: Job-Order Costing

Chapter Three

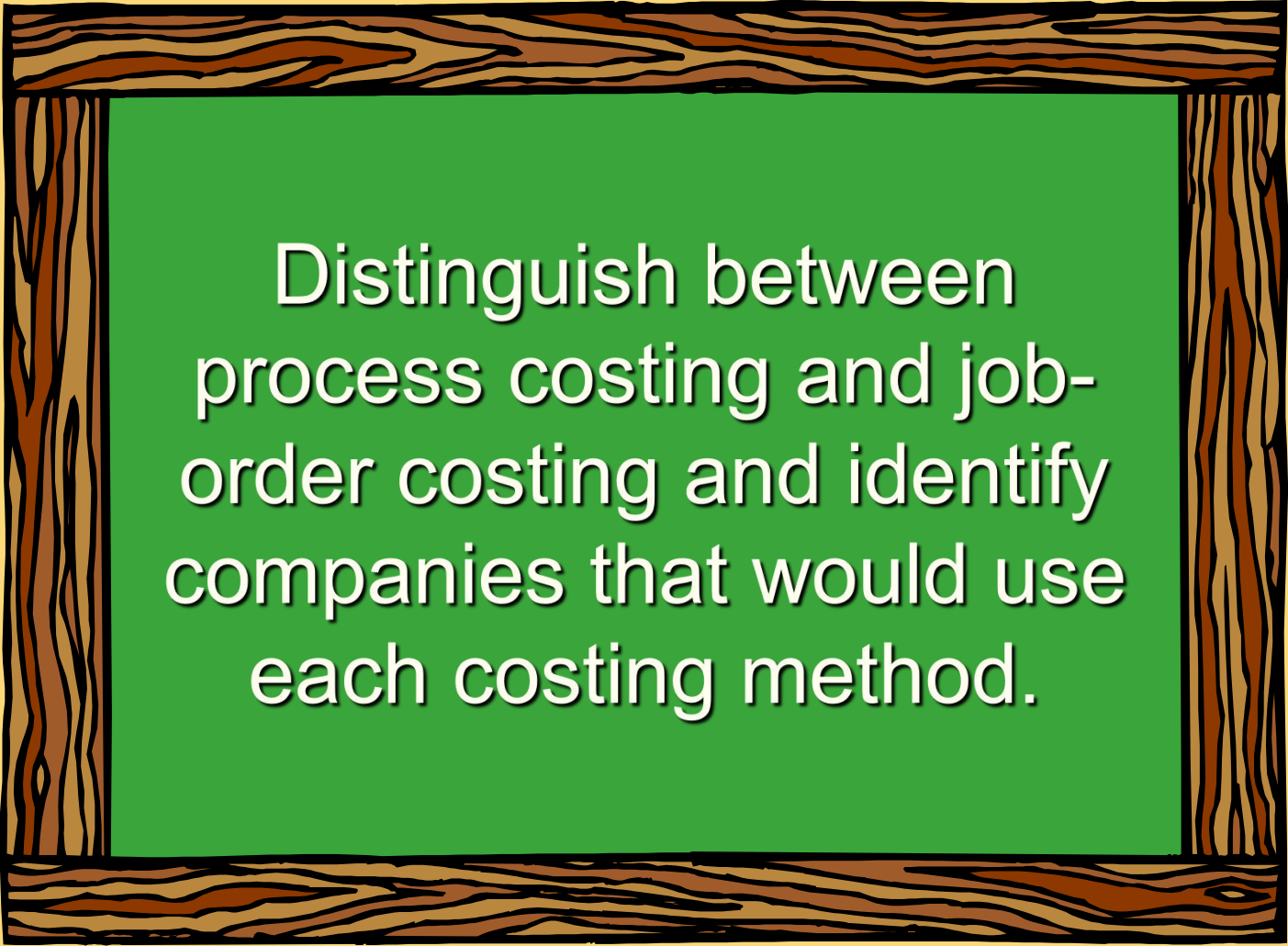


Product Cost Flows

<u>Raw Materials</u>	<u>Manufacturing Costs</u>	<u>Work In Process</u>
Beginning raw materials inventory	Direct materials	Beginning work in process inventory
+ Raw materials purchased	+ Direct labor	+ Total manufacturing costs
<u>= Raw materials available for use in production</u>	<u>+ <u>Mfg. overhead</u></u>	<u>= Total work in process for the period</u>
- Ending raw materials	<u>= <u>Total manufacturing costs</u></u>	- Ending work in process inventory
Costs associated with the goods that are completed during the period are transferred to finished goods inventory.		<u>= Cost of goods manufactured</u>

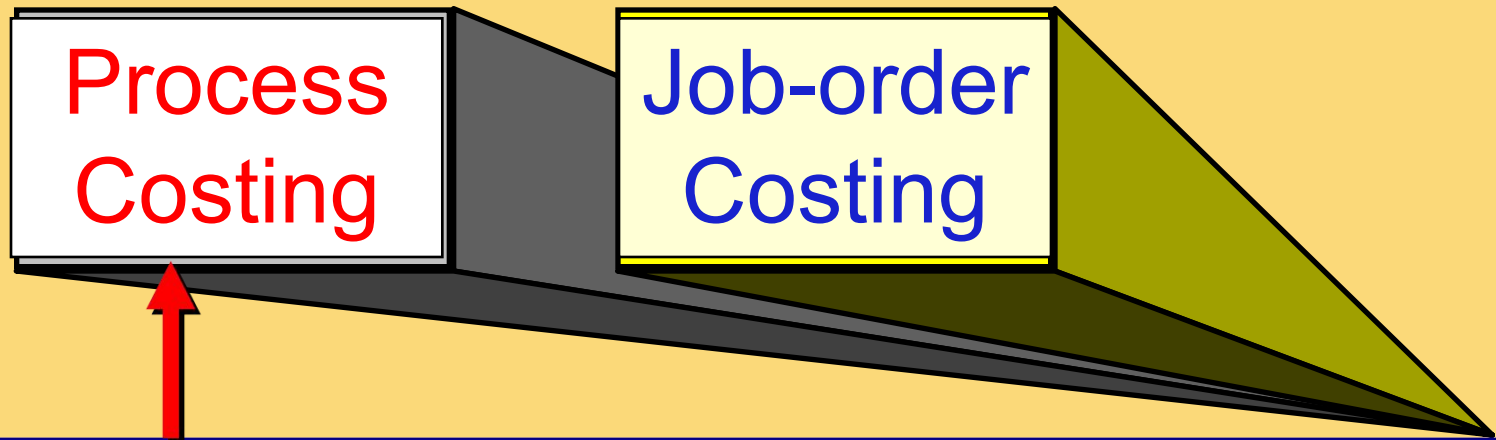


Learning Objective 1



Distinguish between process costing and job-order costing and identify companies that would use each costing method.

Types of Product Costing Systems



- ❖ A company produces many units of a single product.
- ❖ One unit of product is indistinguishable from other units of product.
- ❖ The identical nature of each unit of product enables assigning the same average cost per unit.

Types of Product Costing Systems

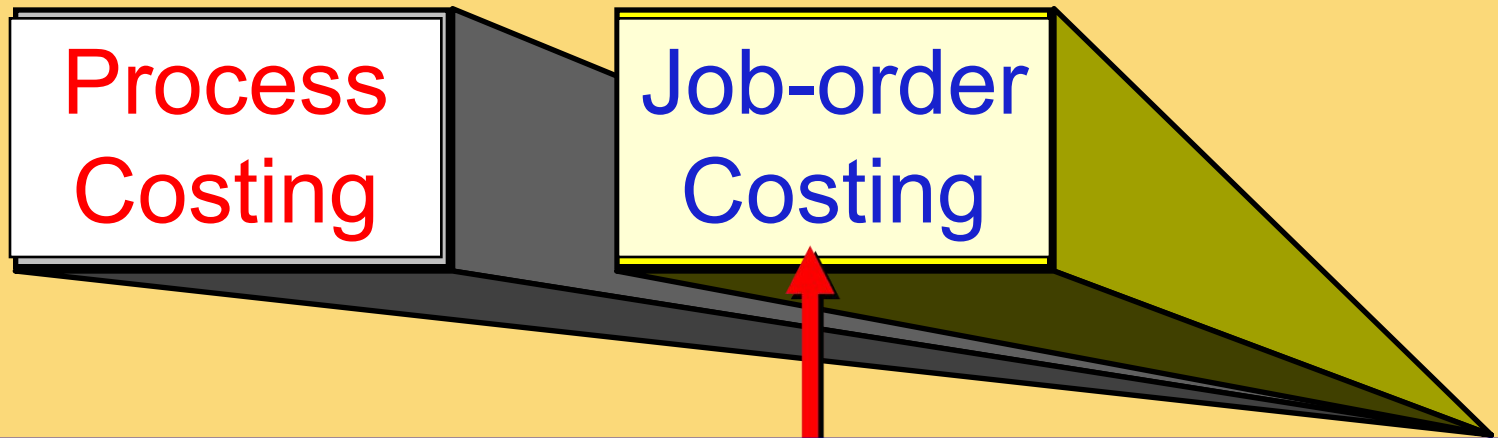


❖ **A company produces many units of a single**

Example companies:

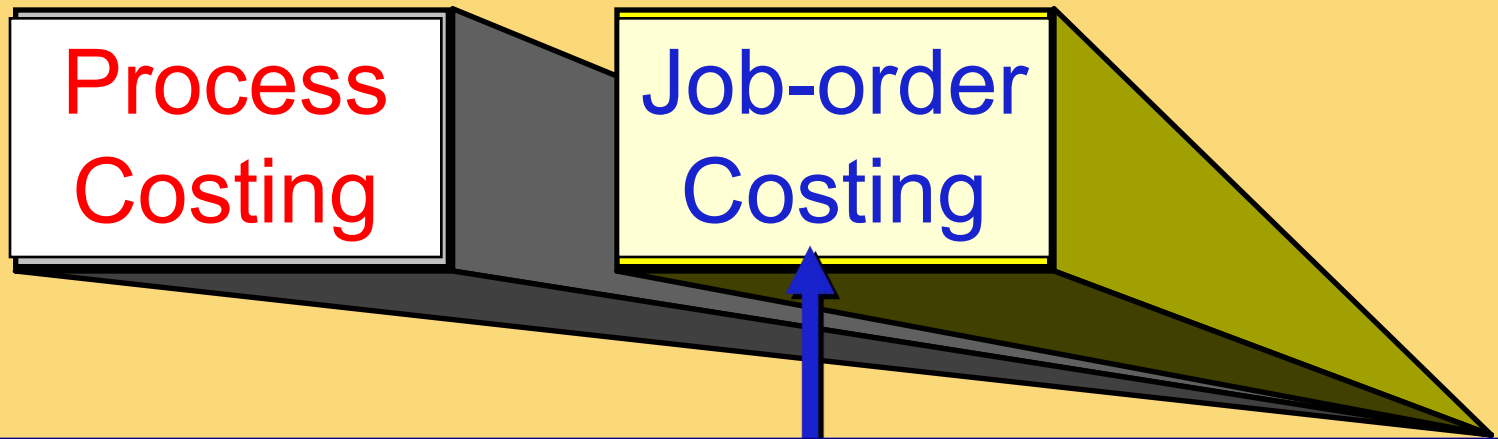
- ❖ 1. Weyerhaeuser (paper manufacturing)
 - ❖ 2. Reynolds Aluminum (refining aluminum ingots)
 - ❖ 3. Coca-Cola (mixing and bottling beverages)
- es
assigning the same average cost per unit.

Types of Product Costing Systems



- ❖ Many different products are produced each period.
- ❖ Products are manufactured to order.
- ❖ The unique nature of each order requires tracing or allocating costs to each job, and maintaining cost records for each job.

Types of Product Costing Systems



Example companies:

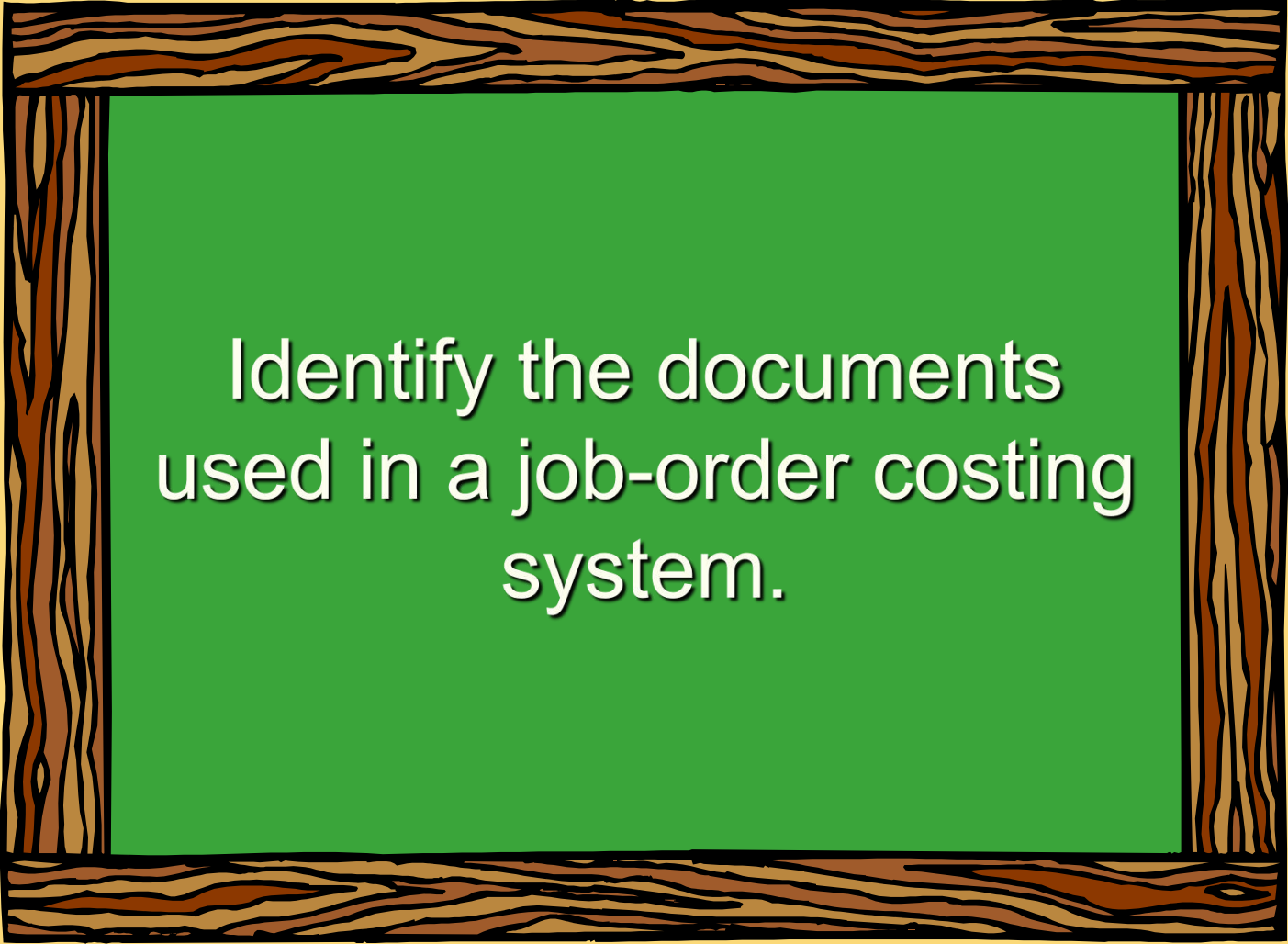
- ❖ 1. Boeing (aircraft manufacturing)
- ❖ 2. Bechtel International (large scale construction)
- ❖ 3. Walt Disney Studios (movie production)

Comparing Process and Job-Order Costing

	Job-Order	Process
Number of jobs worked	Many	Single Product
Cost accumulated by	Job	Department
Average cost computed by	Job	Department

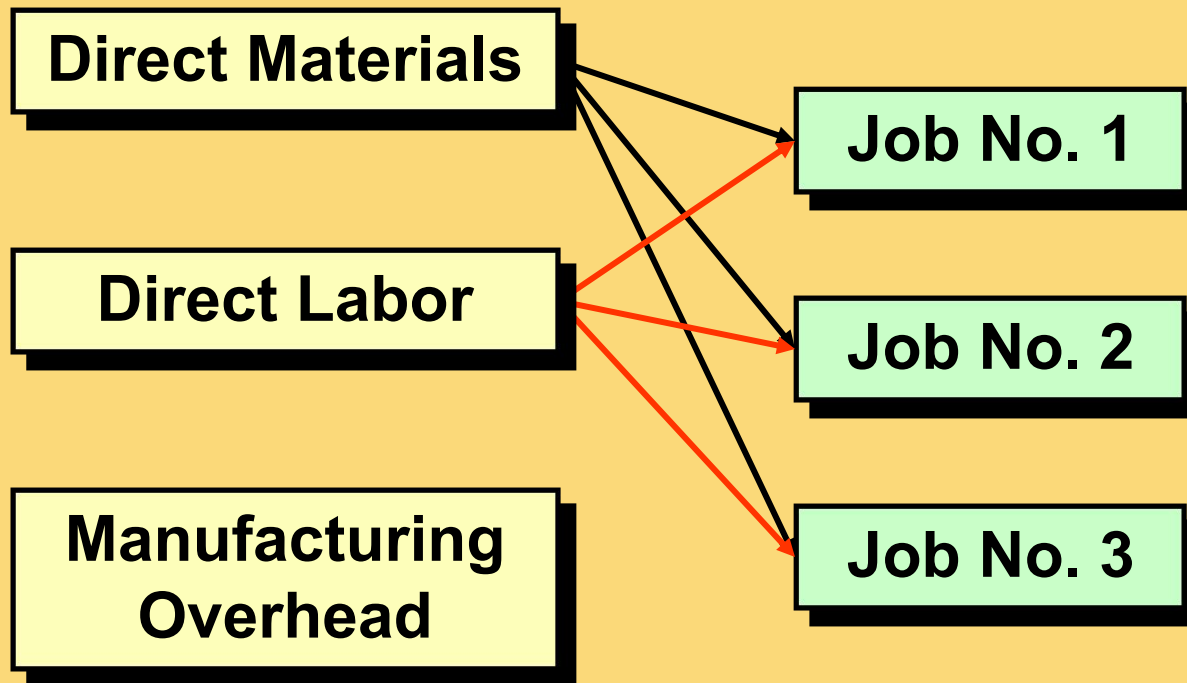


Learning Objective 2



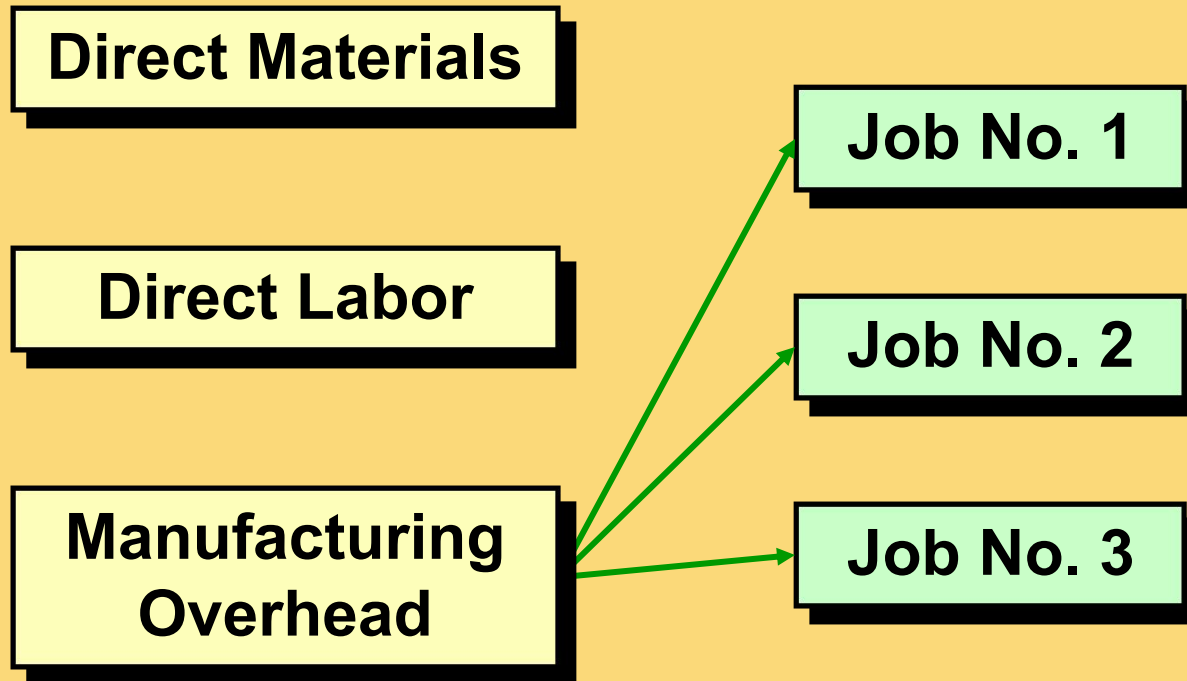
Identify the documents
used in a job-order costing
system.

Job-Order Costing – An Overview



**Charge
direct
material and
direct labor
costs to
each job as
work is
performed.**

Direct Manufacturing Costs



Manufacturing Overhead, including *indirect materials* and *indirect labor*, are allocated to all jobs rather than directly traced to each job.

The Job Cost Sheet



PearCo Job Cost Sheet

Job Number A - 143

Date Initiated 3-4-05

Department B3

Date Completed _____

Item Wooden cargo crate

Units Completed _____

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount

Cost Summary		Units Shipped		
Direct Materials		Date	Number	Balance
Direct Labor				
Manufacturing Overhead				
Total Cost				
Unit Product Cost				

Measuring Direct Materials Cost



PearCo Materials Requisition Form

Requisition No. X7 - 6890

Date 3-4-05

Job No. A - 143

Department B3

Description	Quantity	Unit Cost	Total Cost
2 x 4, 12 feet	12	\$ 3.00	\$ 36.00
1 x 6, 12 feet	20	4.00	80.00
			\$ 116.00

Authorized Will E. Delite
Signature

Measuring Direct Materials Cost



PearCo Job Cost Sheet

Job Number A - 143

Date Initiated 3-4-05

Department B3

Date Completed _____

Item Wooden cargo crate

Units Completed _____

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount
X7-6890	\$ 116						

Cost Summary		Units Shipped		
Direct Materials	\$ 116	Date	Number	Balance
Direct Labor				
Manufacturing Overhead				
Total Cost				
Unit Product Cost				

Measuring Direct Labor Costs



PearCo Employee Time Ticket

Time Ticket No. 36

Date 3/5/2005

Employee I. M. Skilled

Station 42

Starting Time	Ending Time	Hours Completed	Hourly Rate	Amount	Job No.
0800	1600	8.00	\$ 11.00	\$ 88.00	A-143
Totals		8.00	\$ 11.00	\$ 88.00	A-143

Supervisor C. M. Workman

Job-Order Cost Accounting



PearCo Job Cost Sheet

Job Number A - 143

Date Initiated 3-4-05

Department B3

Date Completed _____

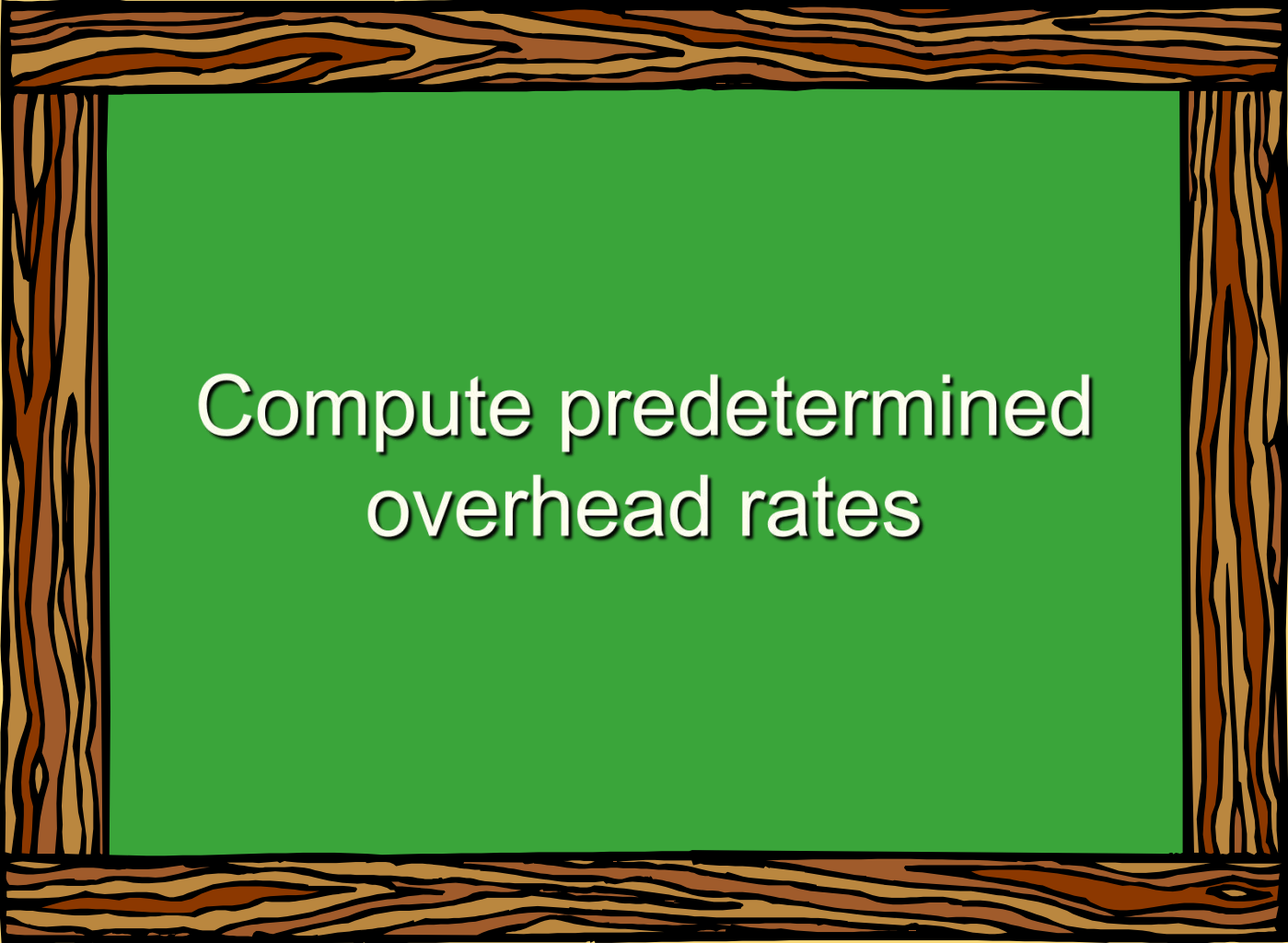
Item Wooden cargo crate

Units Completed _____

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount
X7-6890	\$ 116	36	8	\$ 88			

Cost Summary		Units Shipped		
Direct Materials	\$ 116	Date	Number	Balance
Direct Labor	\$ 88			
Manufacturing Overhead				
Total Cost				
Unit Product Cost				

Learning Objective 3



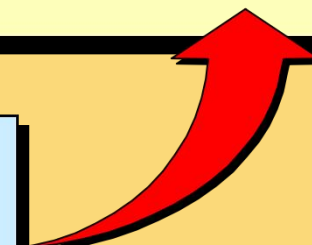
Compute predetermined
overhead rates

Manufacturing Overhead Application

The predetermined overhead rate (*POHR*) used to apply overhead to jobs is determined before the period begins.

$$\text{POHR} = \frac{\text{Estimated total manufacturing overhead cost for the coming period}}{\text{Estimated total units in the allocation base for the coming period}}$$

Ideally, the allocation base is a cost driver that causes overhead.



Application of Manufacturing Overhead

Based on *estimates*, and determined before the period begins.



Overhead applied = POHR × Actual
activity

Actual amount of the allocation based upon the actual level of activity.

Overhead Application Rate

$$\text{POHR} = \frac{\text{Estimated total manufacturing overhead cost for the coming period}}{\text{Estimated total units in the allocation base for the coming period}}$$

$$\text{POHR} = \frac{\$640,000}{160,000 \text{ direct labor hours (DLH)}}$$

$$\text{POHR} = \$4.00 \text{ per DLH}$$

For each direct labor hour worked on a particular job, \$4.00 of factory overhead will be applied to that job.



Job-Order Cost Accounting



PearCo Job Cost Sheet

Job Number A - 143

Date Initiated 3-4-05

Date Completed 3-5-05

Department B3

Units Completed 2

Item Wooden cargo crate

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount
X7-6890	\$ 116	36	8	\$ 88	8	\$ 4	\$ 32

Cost Summary			Units Shipped		
			Date	Number	Balance
Direct Materials	\$	116			
Direct Labor	\$	88			
Manufacturing Overhead	\$	32			
Total Cost					
Unit Product Cost					

Job-Order Cost Accounting



PearCo Job Cost Sheet

Job Number A - 143

Date Initiated 3-4-05

Date Completed 3-5-05

Department B3

Units Completed 2

Item Wooden cargo crate

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount
X7-6890	\$ 116	36	8	\$ 88	8	\$ 4	\$ 32

Cost Summary				Units Shipped		
				Date	Number	Balance
Direct Materials						
Direct Labor						
Manufacturing Overhead						
Total Cost						
Unit Product Cost						

Interpreting the Average Unit Cost

The average unit cost should not be interpreted as the costs that would actually be incurred if an additional unit were produced.

Fixed overhead would not change if another unit were produced, so the incremental cost of another unit may be somewhat less than \$118.

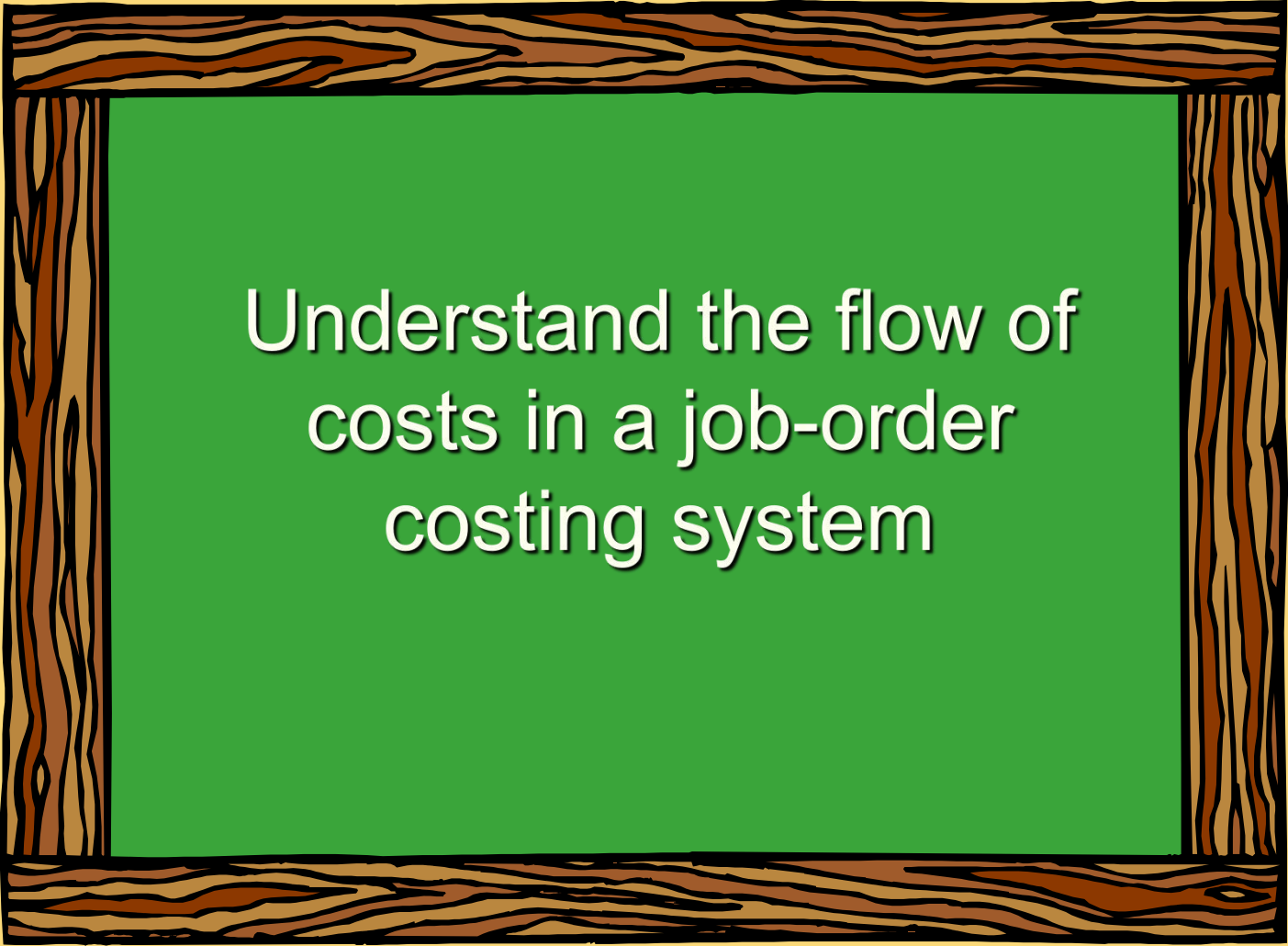
Quick Check ✓



Job WR53 at NW Fab, Inc. required \$200 of direct materials and 10 direct labor hours at \$15 per hour. Estimated total overhead for the year was \$760,000 and estimated direct labor hours were 20,000. What would be recorded as the cost of job WR53?

- a. \$200.
- b. \$350.
- c. \$380.
- d. \$730.

Learning Objective 4



Understand the flow of
costs in a job-order
costing system

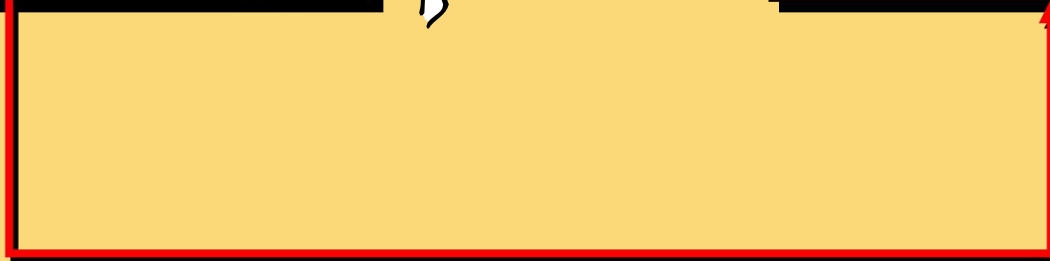
Job-Order Costing

Document Flow Summary

A sales order is the basis of issuing a production order.

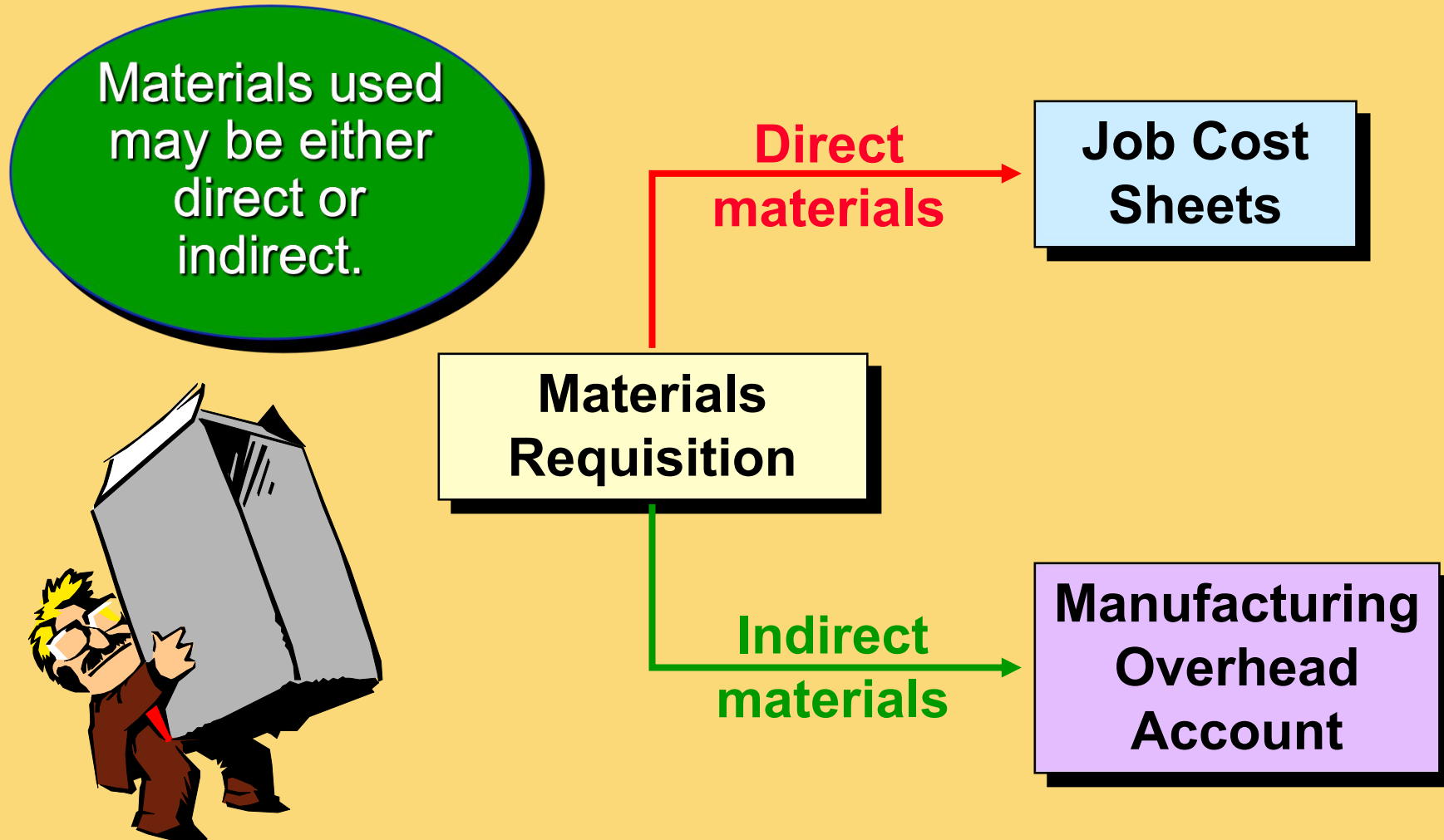


A production order initiates work on a job.



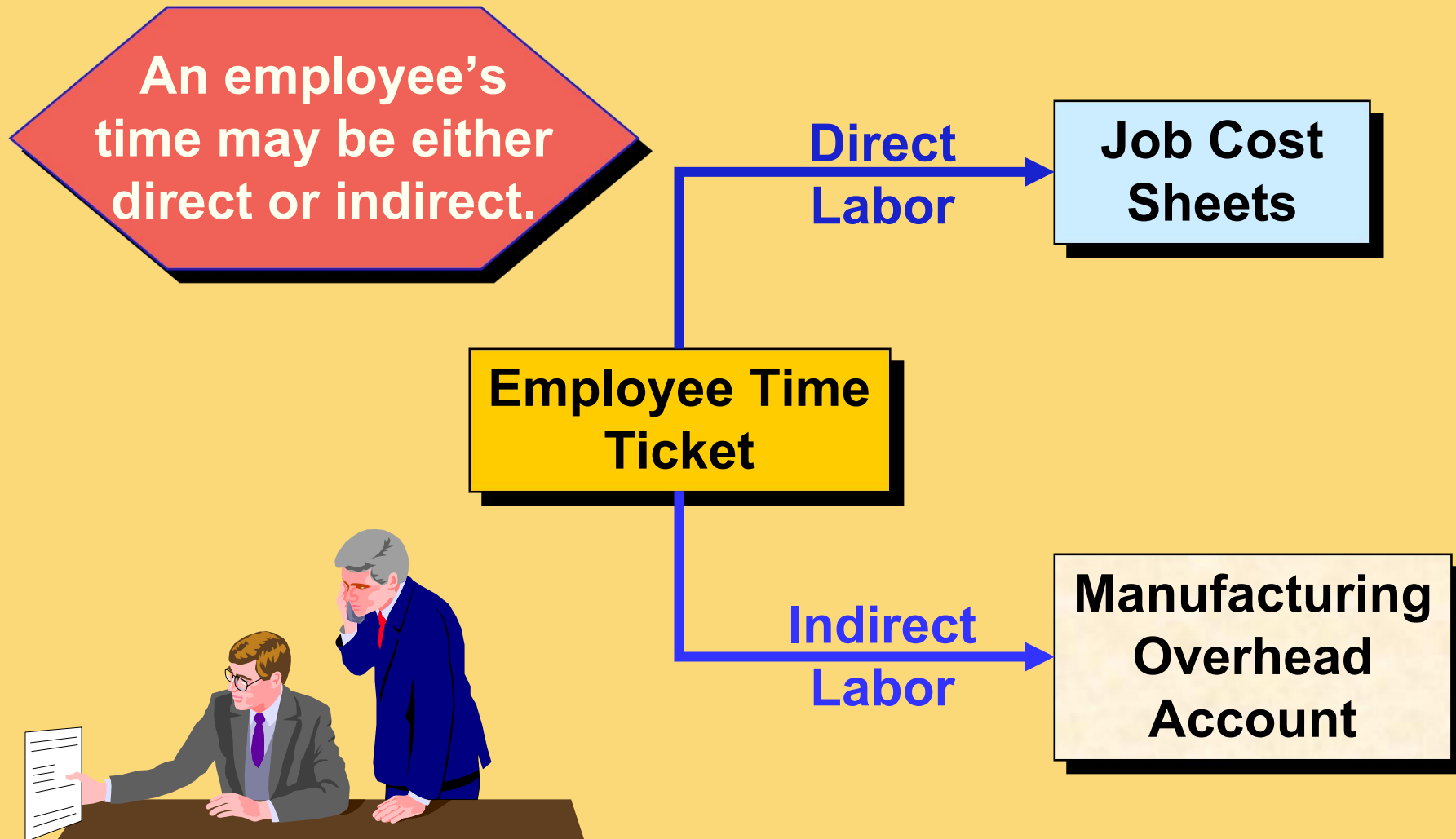
Job-Order Costing

Document Flow Summary



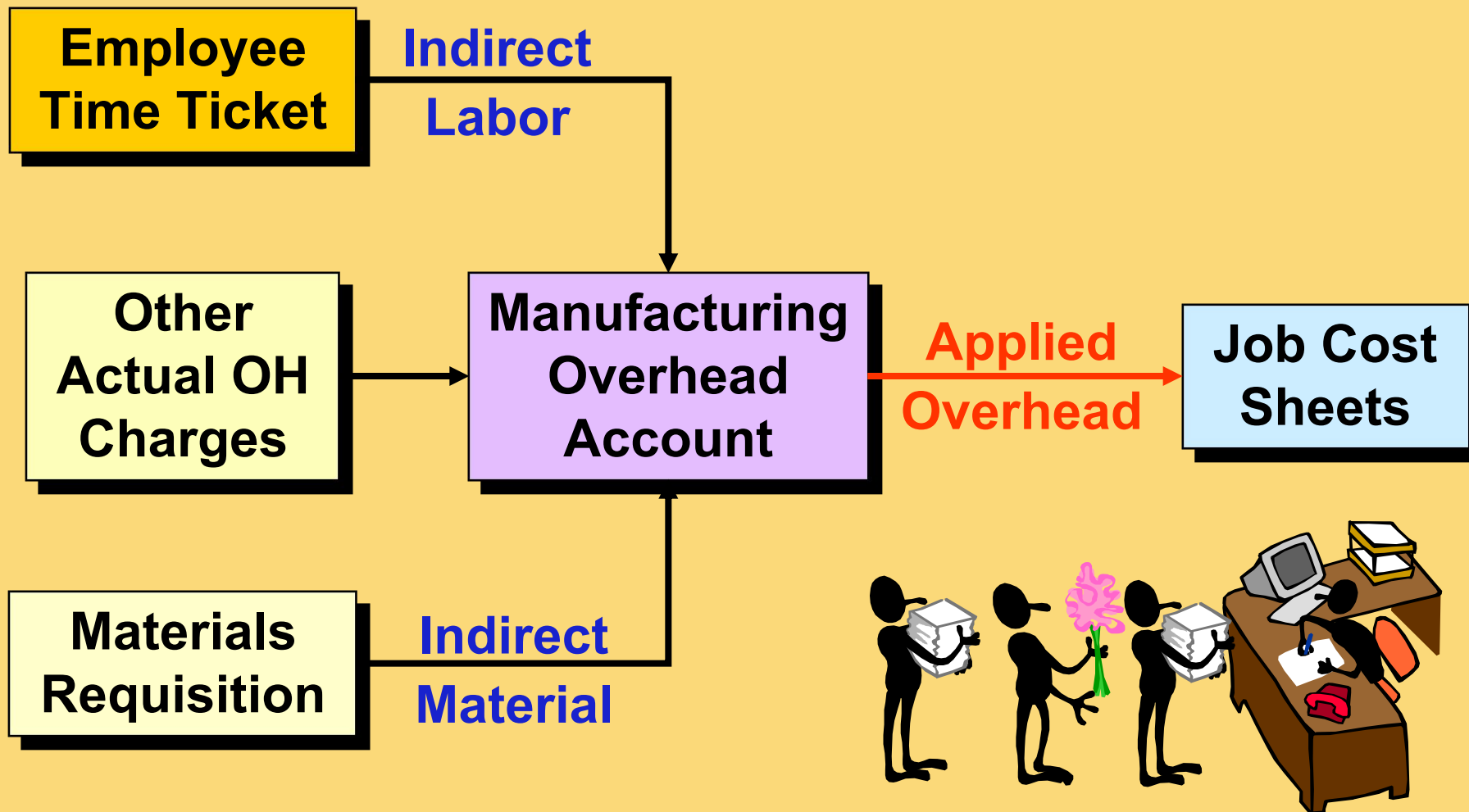
Job-Order Costing

Document Flow Summary

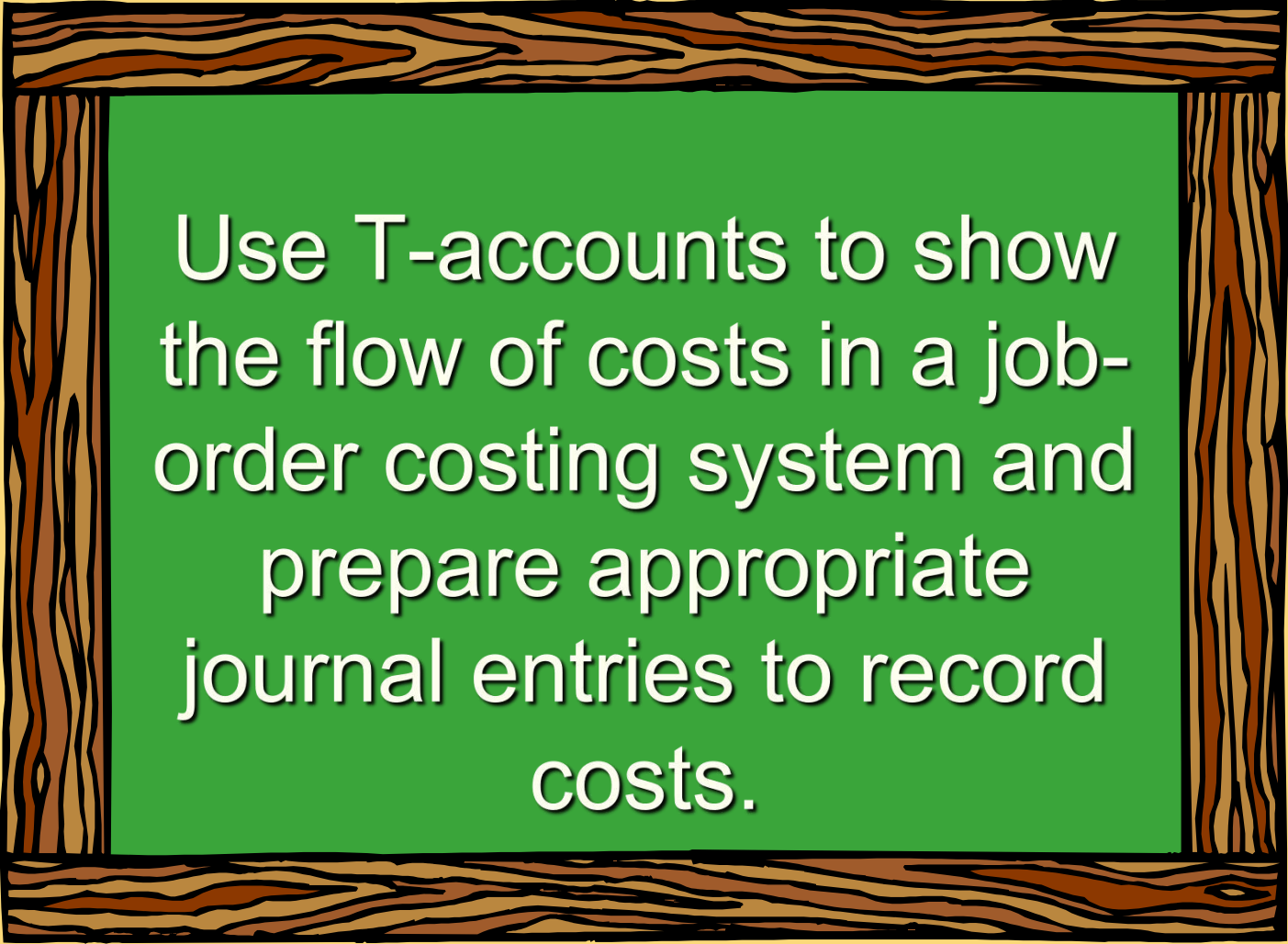


Job-Order Costing

Document Flow Summary

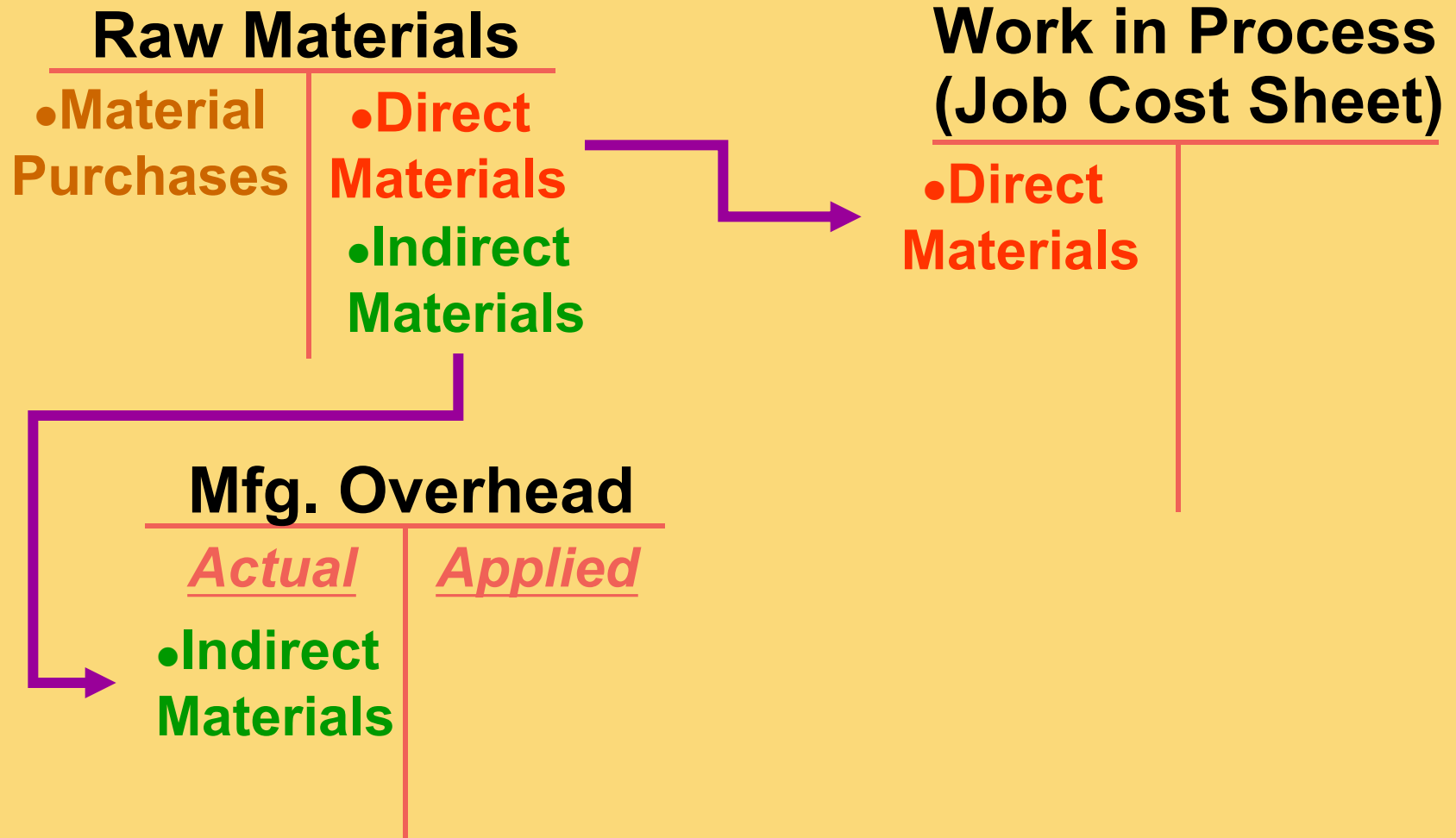


Learning Objectives 5



Use T-accounts to show the flow of costs in a job-order costing system and prepare appropriate journal entries to record costs.

The Purchase and Issue of Raw Materials



Cost Flows – Material Purchases

Raw material purchases are recorded in an inventory account.

GENERAL JOURNAL

Date		Description	Post. Ref.	Debit	Credit
		Raw Materials		XXXXX	
		Accounts Payable			XXXXX

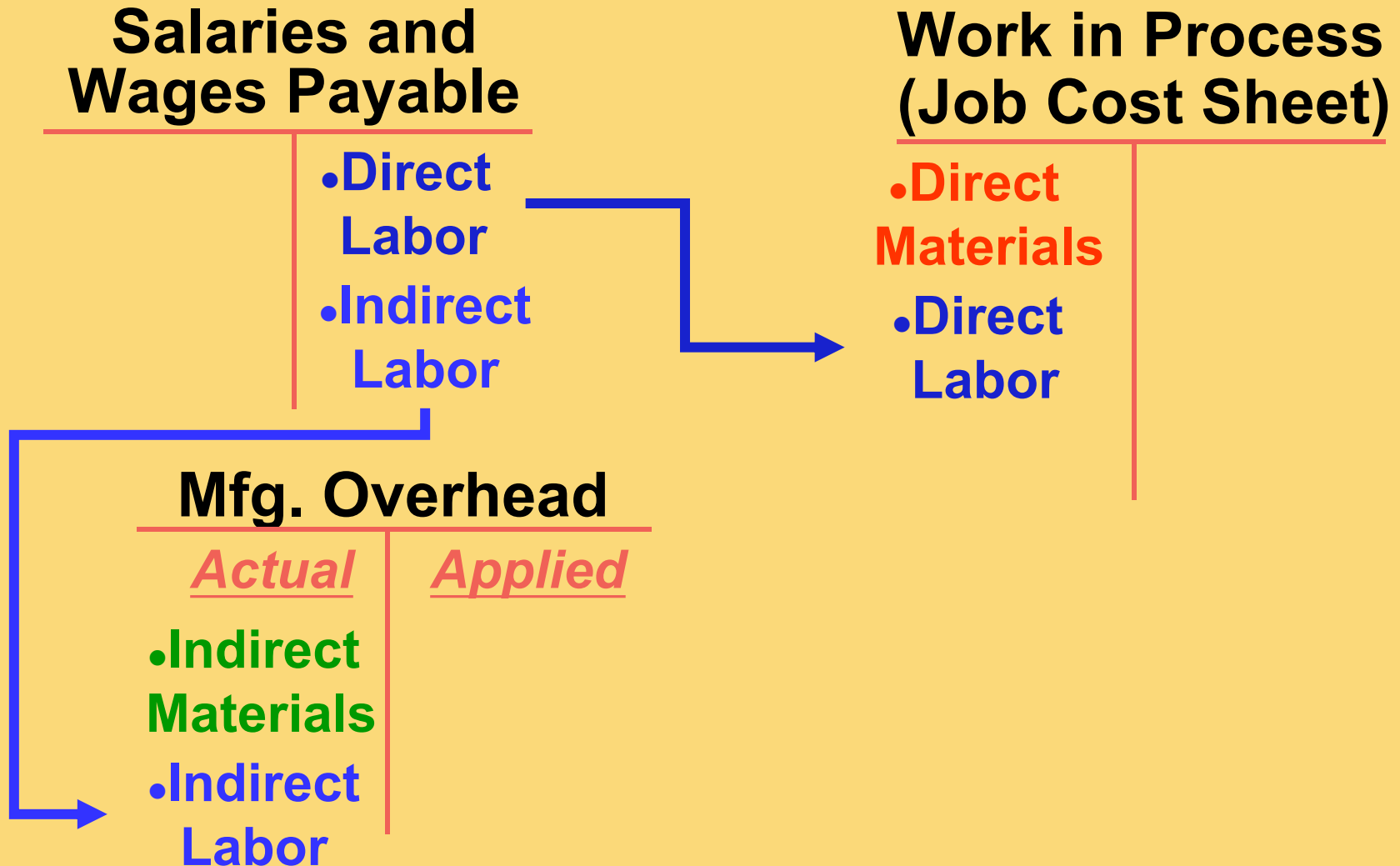
Cost Flows – Material Usage

Direct materials issued to a job increase Work in Process and decrease Raw Materials. Indirect materials used are charged to Manufacturing Overhead and also decrease Raw Materials.

GENERAL JOURNAL

Date		Description	Post. Ref.	Debit	Credit
		Work in Process		XXXXXX	
		Manufacturing Overhead		XXXXXX	
		Raw Materials			XXXXXX

The Recording of Labor Costs



The Recording of Labor Costs

The cost of direct labor incurred increases Work in Process and the cost of indirect labor increases Manufacturing Overhead.

GENERAL JOURNAL

Date		Description	Post. Ref.	Debit	Credit
		Work in Process		XXXXX	
		Manufacturing Overhead		XXXXX	
		Salaries and Wages Payable			XXXXX

Recording Actual Manufacturing Overhead

Salaries and Wages Payable

- Direct Labor
- Indirect Labor

Work in Process (Job Cost Sheet)

- Direct Materials
- Direct Labor

Mfg. Overhead

- Actual
- Indirect Materials
 - Indirect Labor

Applied

•Other Overhead

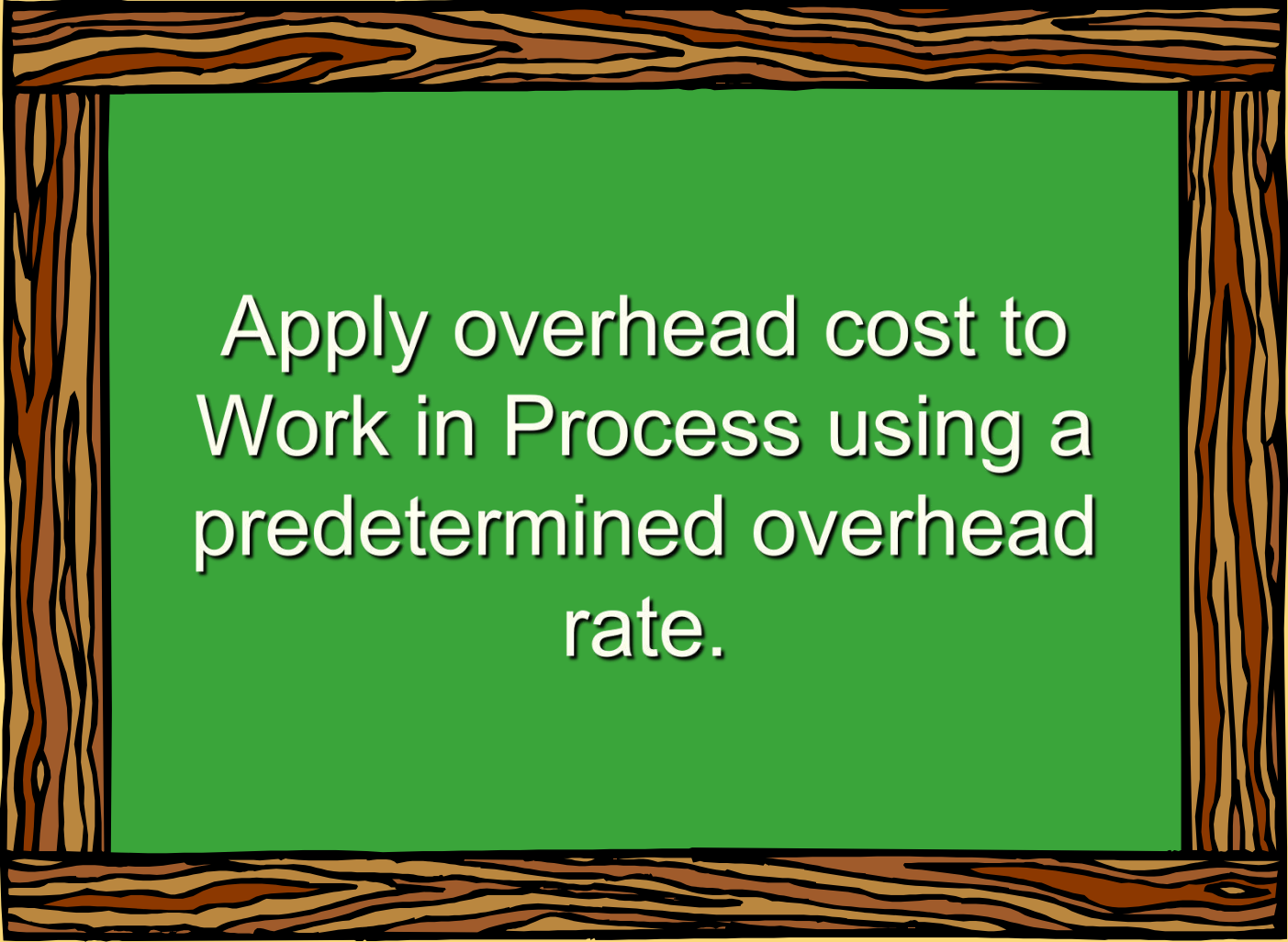
Recording Actual Manufacturing Overhead

In addition to indirect materials and indirect labor, other manufacturing overhead costs are charged to the Manufacturing Overhead account as they are incurred.

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Date		Description	Post. Ref.	Debit	Credit
		Manufacturing Overhead		XXXXX	
		Accounts Payable			XXXXX
		Property Taxes Payable			XXXXX
		Prepaid Insurance			XXXXX
		Accumulated Depreciation			XXXXX

Learning Objective 6



Apply overhead cost to
Work in Process using a
predetermined overhead
rate.

Applying Manufacturing Overhead

Salaries and Wages Payable

- Direct Labor
- Indirect Labor

Mfg. Overhead

<u>Actual</u>	<u>Applied</u>
•Indirect Materials •Indirect Labor •Other Overhead	•Overhead Applied to Work in Process

Work in Process (Job Cost Sheet)

- Direct Materials
- Direct Labor
- Overhead Applied

If actual and applied manufacturing overhead are not equal, a year-end adjustment is required.

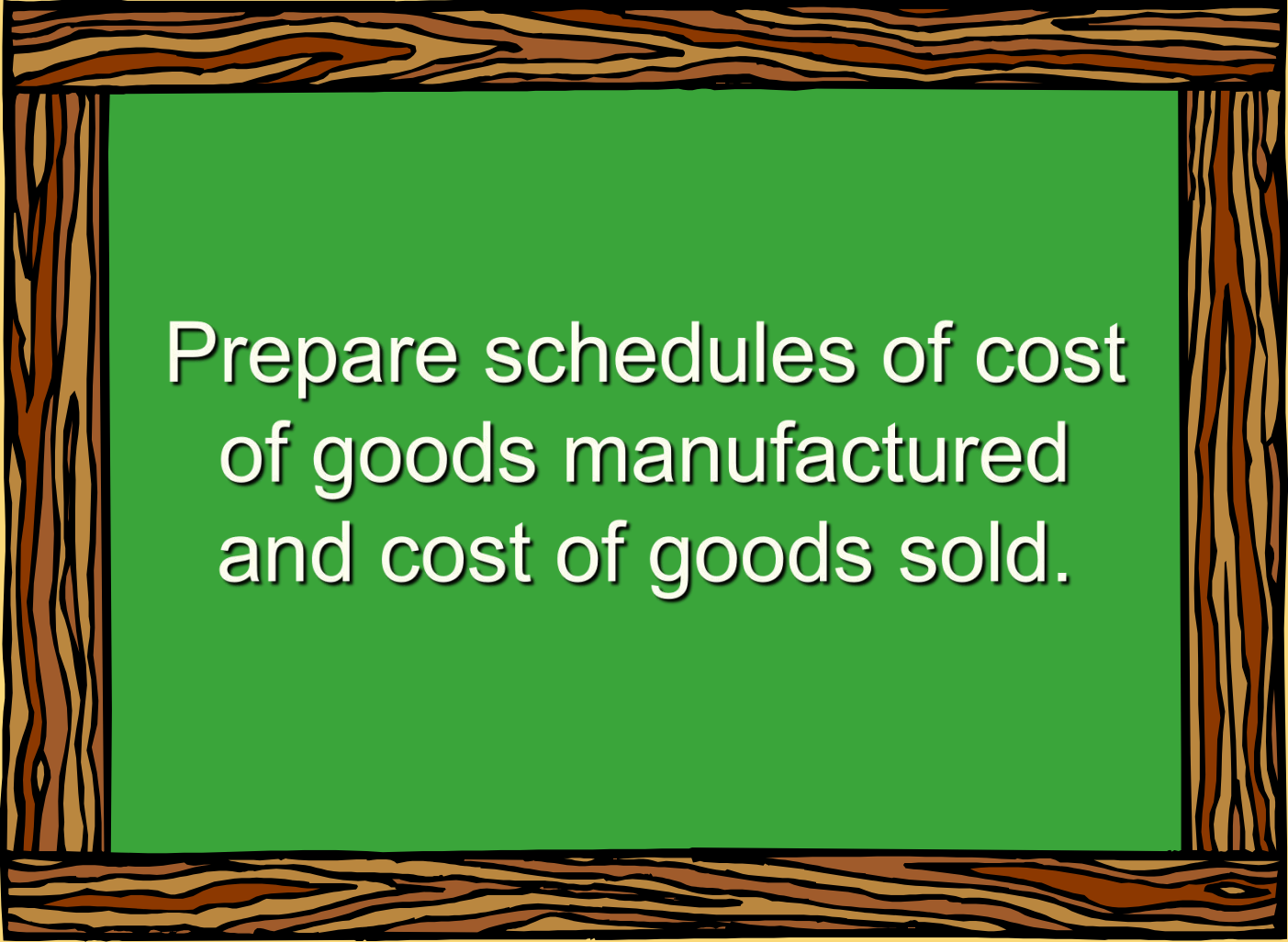
Applying Manufacturing Overhead

Work in Process is increased when Manufacturing Overhead is applied to jobs.

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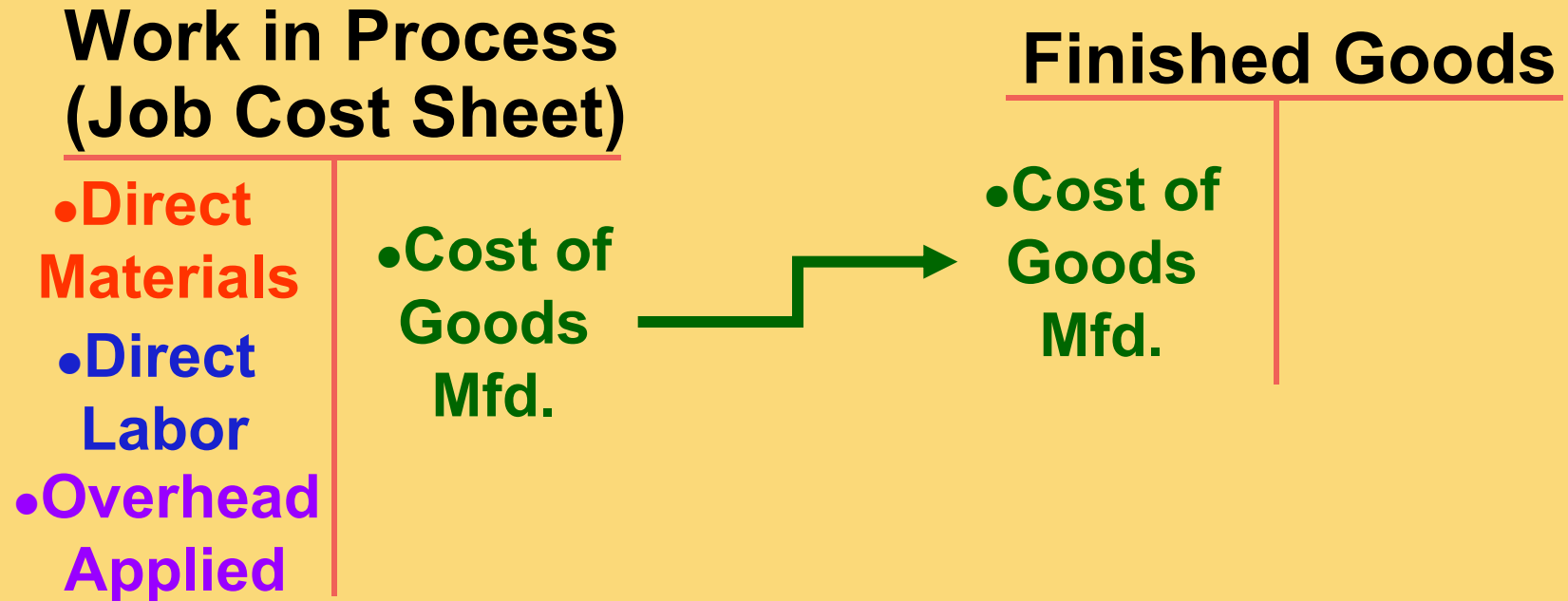
Date		Description	Post. Ref.	Debit	Credit
		Work in Process		XXXXX	
		Manufacturing Overhead			XXXXX

Learning Objective 6



Prepare schedules of cost
of goods manufactured
and cost of goods sold.

Transferring Completed Units



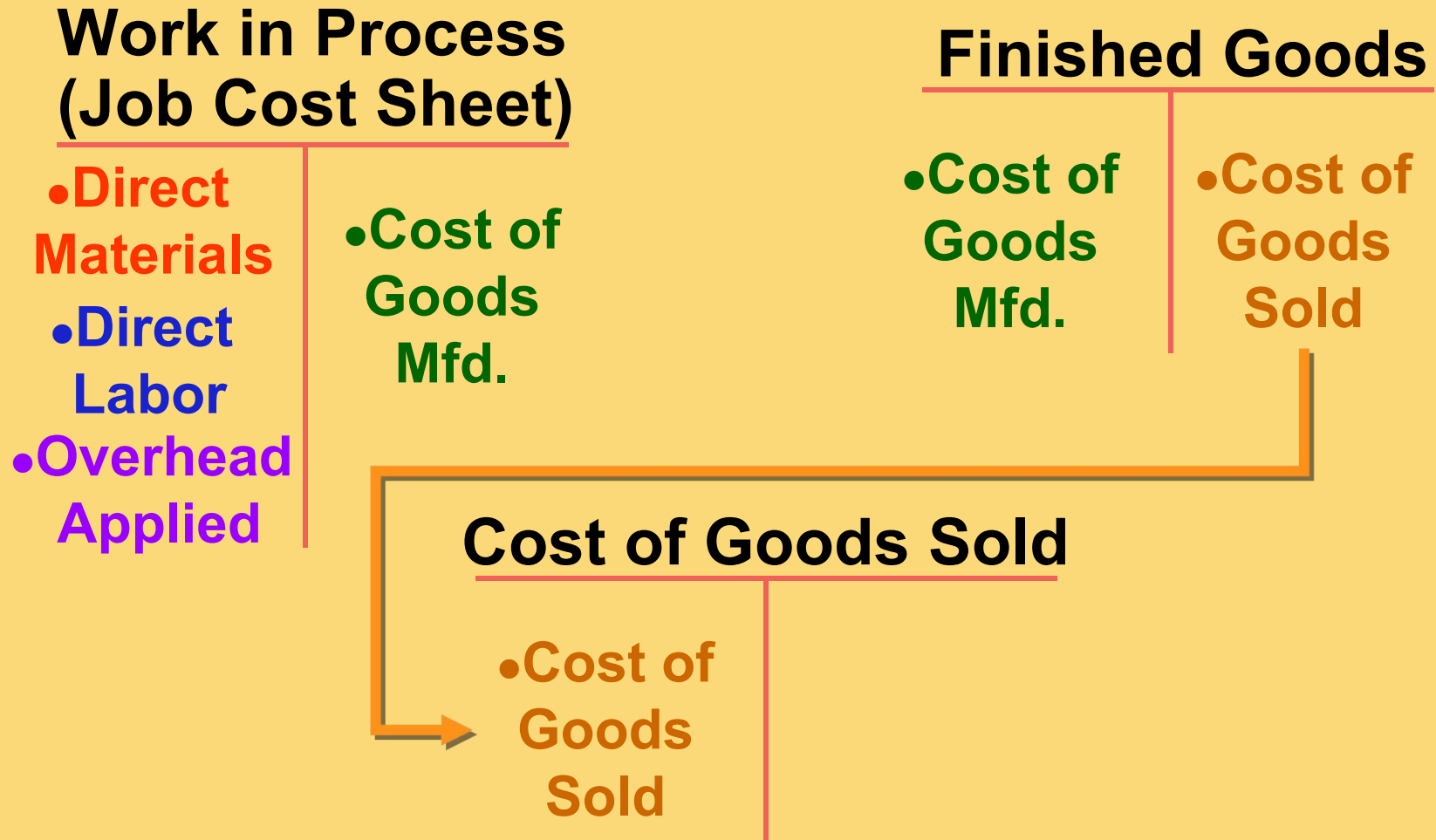
Transferring Completed Units

As jobs are completed, the Cost of Goods Manufactured is transferred to Finished Goods from Work in Process.

GENERAL JOURNAL

Date		Description	Post. Ref.	Debit	Credit
		Finished Goods		XXXXX	
		Work in Process			XXXXX

Transferring Units Sold



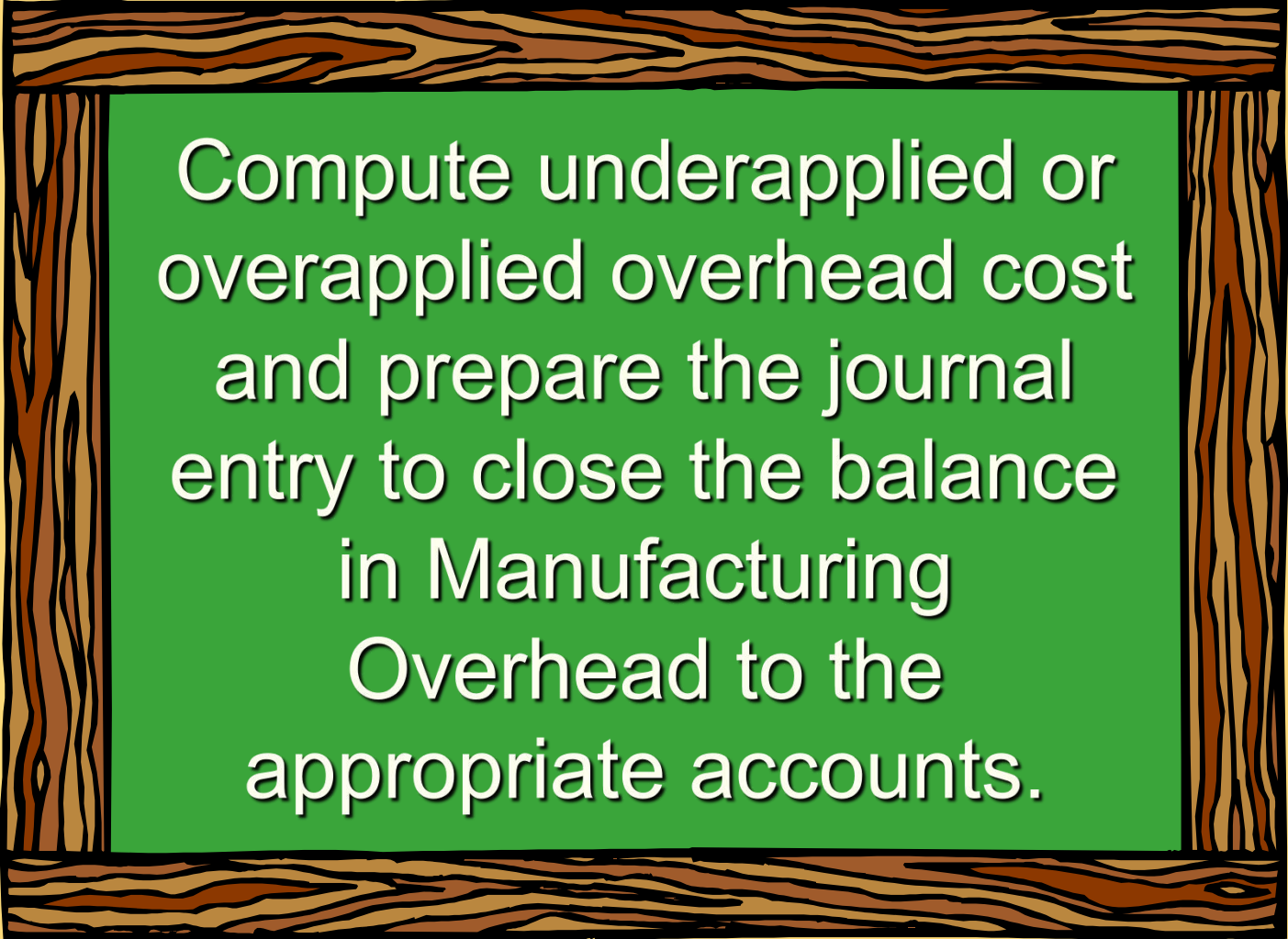
Transferring Units Sold

When finished goods are sold, two entries are required: (1) to record the sale, and (2) to record COGS and reduce Finished Goods.

GENERAL JOURNAL

Date		Description	Post. Ref.	Debit	Credit
		Accounts Receivable		XXXXX	
		Sales			XXXXX
		Cost of Goods Sold		XXXXX	
		Finished Goods			XXXXX

Learning Objective 8



Compute underapplied or overapplied overhead cost and prepare the journal entry to close the balance in Manufacturing Overhead to the appropriate accounts.

Problems of Overhead Application

The difference between the overhead cost applied to Work in Process and the actual overhead costs of a period is referred to as either underapplied or overapplied overhead.

Underapplied overhead exists when the amount of overhead applied to jobs during the period using the predetermined overhead rate is *less than* the total amount of overhead actually incurred during the period.

Overapplied overhead exists when the amount of overhead applied to jobs during the period using the predetermined overhead rate is *greater than* the total amount of overhead actually incurred during the period.

Overhead Application Example



PearCo's *actual overhead* for the year was **\$650,000** with a total of **170,000** direct labor hours worked on jobs.

How much total overhead was applied to PearCo's jobs during the year? Use PearCo's predetermined overhead rate of \$4.00 per direct labor hour.

Overhead Applied During the Period

Applied Overhead = POHR $\times$ Actual Direct Labor Hours

Applied Overhead = \$4.00 per DLH $\times$ 170,000 DLH =
\$680,000

Overhead Application Example



PearCo's *actual overhead* for the year was **\$650,000** with a total of **170,000** direct labor

PearCo has *overapplied* overhead for the year by \$30,000. What will PearCo do?



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? Use
ad rate of
ur

Overhead Applied During the Period

Applied Overhead = POHR × Actual Direct Labor Hours

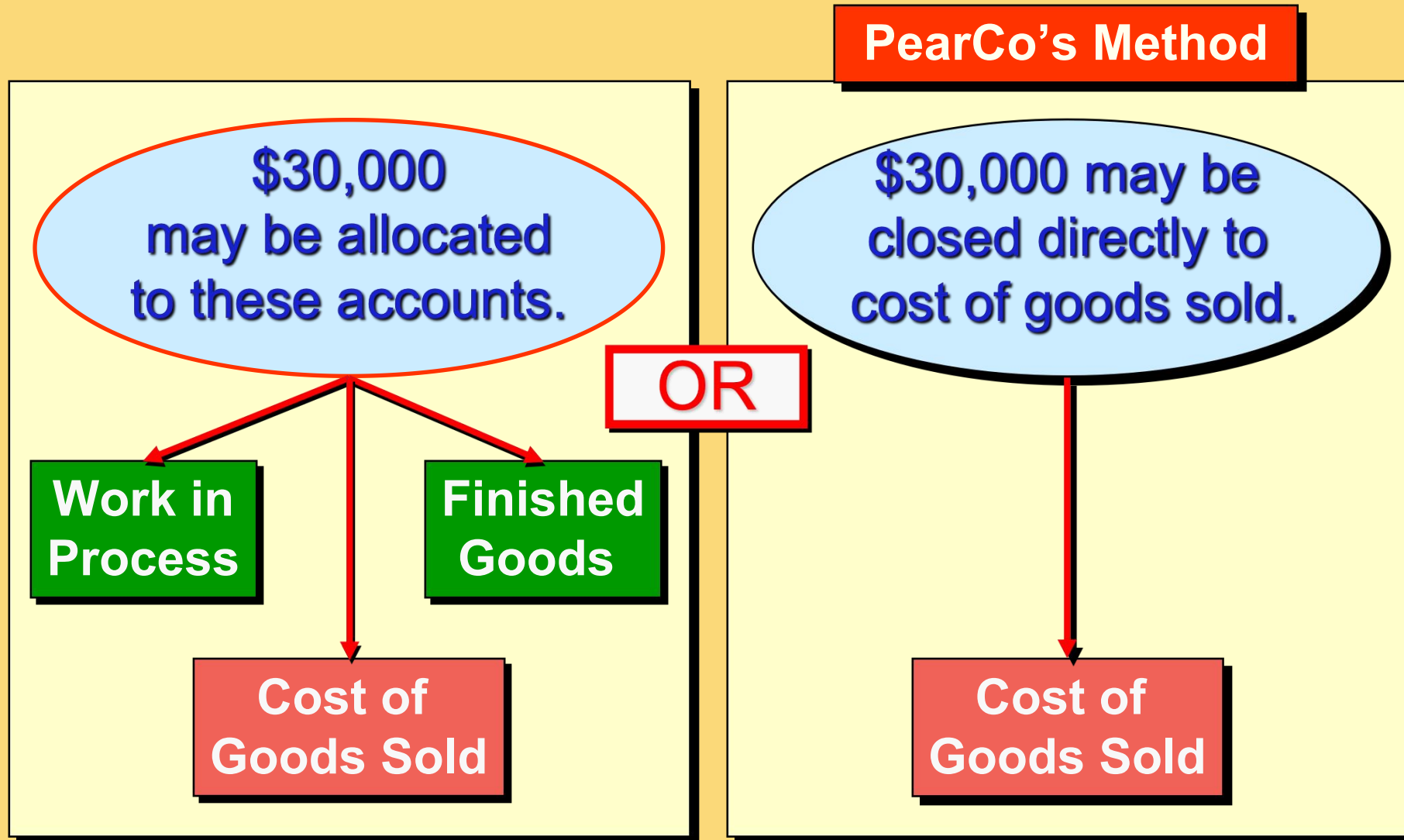
Applied Overhead = \$4.00 per DLH × 170,000 DLH =
\$680,000

Quick Check ✓

Tiger, Inc. had actual manufacturing overhead costs of \$1,210,000 and a predetermined overhead rate of \$4.00 per machine hour. Tiger, Inc. worked 290,000 machine hours during the period. Tiger's manufacturing overhead is

- a. \$50,000 overapplied.**
- b. \$50,000 underapplied.**
- c. \$60,000 overapplied.**
- d. \$60,000 underapplied.**

Disposition of Under- or Overapplied Overhead



Disposition of Under- or Overapplied Overhead

PearCo's Cost of Goods Sold

Unadjusted Balance	
	\$30,000
Adjusted Balance	

PearCo's Mfg. Overhead

Actual overhead costs	Overhead applied to jobs
\$650,000	\$680,000
\$30,000	\$30,000 overapplied



Allocating Under- or Overapplied Overhead Between Accounts

Assume the overhead applied in ending Work in Process Inventory, ending Finished Goods Inventory, and Cost of Goods Sold is shown below:

	Amount
Work in process	\$ 68,000
Finished Goods	204,000
Cost of Goods Sold	408,000
Total	<u>\$ 680,000</u>



Allocating Under- or Overapplied Overhead Between Accounts

We would complete the following allocation of \$30,000 overapplied overhead:

	Amount	Percent of Total	Allocation of \$30,000
Work in process	\$ 68,000	10%	\$ 3,000
Finished Goods	204,000	30%	9,000
Cost of Goods Sold	408,000	60%	18,000
Total	<u>\$ 680,000</u>	<u>100%</u>	<u>\$ 30,000</u>



Allocating Under- or Overapplied Overhead Between Accounts

	Amount	Percent of Total	Allocation of \$30,000
Work in process	\$ 68,000	10%	\$ 3,000
Finished Goods	204,000	30%	9,000
Cost of Goods Sold	408,000	60%	18,000
Total	<u>\$ 680,000</u>	<u>100%</u>	<u>\$ 30,000</u>

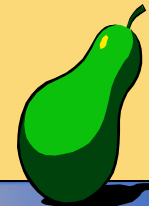
GENERAL JOURNAL

Date	Description	Post. Ref.	Debit	Credit
	Manufacturing Overhead		30,000	
	Work in Process Inventory			3,000
	Finished Goods Inventory			9,000
	Cost of Goods Sold			18,000

Overapplied and Underapplied Manufacturing Overhead - Summary

	PearCo's Method	
If Manufacturing Overhead is . . .	<u>Alternative 1</u> Close to Cost of Goods Sold	<u>Alternative 2</u> Allocation
UNDERAPPLIED (Applied OH is less than actual OH)	INCREASE Cost of Goods Sold	INCREASE Work in Process Finished Goods Cost of Goods Sold
OVERAPPLIED (Applied OH is greater than actual OH)	DECREASE Cost of Goods Sold	DECREASE Work in Process Finished Goods Cost of Goods Sold

Quick Check ✓



What effect will the overapplied overhead have on PearCo's net operating income?

- a. Net operating income will increase.
- b. Net operating income will be unaffected.
- c. Net operating income will decrease.



End of Chapter 3

